Xihong Su

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Education

University of New Hampshire Durham, NH
Ph.D. in Computer Science
Advisor : Marek Petrik
Dissertation: Efficient Algorithms for Mitigating Uncertainty and Risk in Reinforcement Learning

Harbin Institute of Technology Harbin, China
Ph.D. in Computer Architecture
Advisor : Xiaozong Yang
Dissertation: Research on Software Deployment Strategy and Reliability in Distributed System

Harbin Institute of Technology Harbin, China
M.E. in Computer Science and Technology
Advisor : Gang Cui
Thesis: Design and Implementation of CAN High Layer Protocol for Automotive Network Control System Based on J1939

Northeast Petroleum University Daqing, China
B.E. in Computer Science and Technology
Advisor : Chunsheng Li
Thesis: Student Performance and Grade Management System

Experience

SUAI (self-employed) | Artificial Intelligence Researcher 4/2026 - present

- Conduct independent research on reinforcement learning, focusing on average-reward and risk-sensitive decision-making under uncertainty.
- Reinforcement Learning and Optimal Control: studied and practiced Python/PyTorch implementations of DQN, PPO, DDPG, TD3, SAC, IQL, CQL, LQR, and MPC.
- LLM Training and Post-Training: explored Transformer architectures and hands-on implementations of SFT, RLHF, GRPO, and RLVR, focusing on attention mechanisms, policy optimization, and reward-based learning.
- NLP and LLM Applications: gained practical experience with encoder-decoder Transformers, text classification, LangChain, and retrieval-augmented generation (RAG).

University of New Hampshire | Graduate RA/TA 1/2020 - 12/2025

- Proposed exponential value iteration, policy iteration, and linear programming algorithms to compute the optimal policy for the risk-averse total undiscounted reward criterion, under the Entropic Risk Measure (ERM) and Entropic Value at Risk (EVaR) measures, to mitigate risks, optimize performance, and minimize the probability of catastrophic outcomes. The computed optimal strategies are interpretable and practical to deploy.
- Proposed the first risk-averse Q-learning algorithms for the total undiscounted reward ERM and EVaR objectives, established rigorous convergence guarantees to the optimal risk-averse value functions, and validated fast, stable convergence empirically. The proposed model-free gradient-based method enables scalable approximate Q-learning with deep neural networks.
- Derived the policy gradient of Multi-model Markov decision processes (MMDPs) and proposed coordinate ascent dynamic programming (CADP) algorithm to adjust model weights to make robust decisions and improve stability under distribution shift.
- Assisted with instruction, grading, office hours, and student support for courses including: Machine Learning (Python, R), Algorithms (C), Computer Graphics (C++), Database Systems(SQL), Introduction to Computer Science II(Java).

Harbin Institute of Technology | Graduate RA 9/2007 - 12/2011

- Developed optimization and reliability models for component-based distributed software deployment in dynamic environments.
- Designed approximate and heuristic algorithms for reliability-aware software redeployment and component replication under resource constraints.
- Modeled component failures and fault propagation using probabilistic and stochastic methods, including NHPP-based reliability models.
- Analyzed how deployment architecture, component interactions, execution frequency, and replication strategies affect system reliability and availability.

Projects

LLM Prompt Optimization via Reward Feedback

- Built a lightweight LLM-based decision system to generate candidate responses from prompts.
- Designed reward functions to evaluate outputs and select higher-reward responses.
- Simulated policy improvement through iterative reward-based selection.

Robust Reinforcement Learning for Sim-to-Real Control

- Developed continuous-control reinforcement learning policies in simulation.
- Modeled sim-to-real gaps using structured variations in system dynamics.
- Applied domain randomization and evaluated policy robustness under perturbed environments.

Publications

1. **Xihong Su**, Jia Lin Hau, Gersi Doko, Kishan Panaganti, Marek Petrik. Risk-Averse Total-Reward Reinforcement Learning. The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).
2. **Xihong Su**, Marek Petrik, Julien Grand-Clément. Risk-averse Total-reward MDPs with ERM and EVaR. The 39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025) (Oral, 4.6%).
3. **Xihong Su**, Marek Petrik, Julien Grand-Clément. Optimality of Stationary Policies in Risk-averse Total-reward MDPs with EVaR. ICML 2024 Workshop: Foundations of Reinforcement Learning and Control–Connections and Perspectives.
4. **Xihong Su**, Marek Petrik, Julien Grand-Clément. EVaR Optimization in MDPs with Total Reward Criterion. Seventeenth European Workshop on Reinforcement Learning (EWRL17 2024).
5. **Xihong Su**, Marek Petrik. Solving Multi-model MDPs by Coordinate Ascent and Dynamic Programming. Uncertainty in Artificial Intelligence (UAI), 2023.
6. **Xihong Su**, Zhibo Wu, Hongwei Liu, Decheng Zuo, Xiaozong Yang. A Comprehensive Approach of SA Based Software Deployment Reliability Estimation in Neural Networks. 2010 First International Conference on Pervasive Computing, Signal Processing and Applications.
7. **Xihong Su**, Zhibo Wu, Hongwei Liu, Xiaozong Yang, XiaoZong Yang, Decheng Zuo. Reliability analysis of SA based software deployment with consideration of system deployment. Intelligent Automation & Soft Computing 17(6): 749-758, 2011.

Skills

`Python`, `Julia`, `C/C++`, `Scala`, `Java`, `R`, `Git`, `PyTorch`, `Matlab`, `SQL`.

Certificate

Generative AI with Large Language Models.

Citizenship

China.